

Midterm 2 Math 2270 Fall 2012

(1) $A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \quad \begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = (1-\lambda)(3-\lambda) - 6 = \lambda^2 - 4\lambda - 5 = (\lambda-5)(\lambda+1)$

$\lambda = 5 \quad N \begin{pmatrix} -4 & 4 \\ 2 & -2 \end{pmatrix} = \langle \begin{pmatrix} 1 \\ 1 \end{pmatrix} \rangle \quad \lambda = -1 \quad N \begin{pmatrix} 2 & 4 \\ 2 & 4 \end{pmatrix} = \langle \begin{pmatrix} -2 \\ 1 \end{pmatrix} \rangle$

$S = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix} \quad S^{-1} = \frac{1}{3} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} \quad S^{-1}AS = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$

(2) $\begin{vmatrix} t & 1 & 0 \\ 1 & t & 1 \\ 0 & 1 & t \end{vmatrix} = t^3 - t - t = t^3 - 2t \quad A^{-1} = \frac{1}{t^3 - 2t} \begin{pmatrix} t^2 - 1 & -t & 1 \\ -t & t^2 - t & 1 \\ 1 & -t & t^2 - 1 \end{pmatrix} \quad A^{-1} = \frac{1}{|A|} C^T$

- (3) ① $\det I = 1$ ② Switching 2 rows changes the sign of the det.
 (3) The determinant is linear in each row.

If two rows are equal, then switching the rows then does not change the matrix. By property 2 we have $\det A = -\det A$ so $2\det A = 0$ so $\det A = 0$

(4) $|A^2| = 4 \quad |2A| = 2^5 |A| = 2^5 \cdot 3 = 96 \quad |A^{-1}| = \frac{1}{3} \quad |A^T A| = |A^T| |A| = 3 \cdot 3 = 9$

$y = mx + b \quad \begin{cases} 3 = 2x + b \\ 5 = 4x + b \\ 2 = -x + b \\ -3 = -5x + b \end{cases} \quad \begin{pmatrix} 2 & 1 \\ 4 & 1 \\ -1 & 1 \\ -5 & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 2 \\ -3 \end{pmatrix}$ Can't solve so multiply by A^T

$\begin{pmatrix} 2 & 4 & -1 & -5 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 4 & 1 \\ -1 & 1 \\ -5 & 1 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 2 & 4 & -1 & -5 \\ 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ 2 \\ -3 \end{pmatrix} \Rightarrow \begin{pmatrix} 4 & 6 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} 39 \\ 7 \end{pmatrix} \Rightarrow m = 39/46, b = 7/4$

(6) a) $V^\perp = \{ \bar{x} \in \mathbb{R}^n \mid \bar{x} \cdot v = 0 \forall v \in V \}$ b) $\dim V^\perp = n - r$

c) $V \subset (V^\perp)^\perp$ because $\forall v \in V$ then $v \cdot \bar{x} = 0 \forall \bar{x} \in V^\perp$ by a)

but then $\dim(V^\perp)^\perp = n - (n - r) = r = \dim V$ so $V = (V^\perp)^\perp$

(7) $N \begin{pmatrix} 1 & 1 & 2 & -1 & -2 \\ 2 & 3 & 1 & 1 & 1 \end{pmatrix} = N \begin{pmatrix} 1 & 2 & -1 & -2 \\ 0 & -1 & 3 & 5 \end{pmatrix} \Rightarrow N \begin{pmatrix} 1 & 0 & 5 & 8 \\ 0 & 1 & -3 & -5 \end{pmatrix} = \langle \begin{pmatrix} -5 \\ 3 \\ 1 \\ 1 \end{pmatrix} \rangle w_1 = \frac{1}{\sqrt{35}} \begin{pmatrix} -5 \\ 3 \\ 1 \\ 1 \end{pmatrix}$
 $u_2 = v_2 - \text{proj}_{w_1} v_2 = \begin{pmatrix} -8 \\ 5 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{35}} \begin{pmatrix} -5 \\ 3 \\ 1 \\ 1 \end{pmatrix} \frac{(1-5 \cdot 3 + 0 \cdot 1 + 0 \cdot 1)}{\sqrt{35}} \begin{pmatrix} -5 \\ 3 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -8 \\ 5 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{35} \begin{pmatrix} -5 \\ 3 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -8 + 5/35 \\ 5 - 3/35 \\ 0 - 1/35 \\ 1 - 1/35 \end{pmatrix} = \begin{pmatrix} -11/7 \\ 2/7 \\ -1/7 \\ 6/7 \end{pmatrix}$
 $w_2 = \left(\frac{\sqrt{175}}{2} \right) \begin{pmatrix} -1/2 \\ 2/7 \\ -1/7 \\ 6/7 \end{pmatrix} = \frac{1}{5\sqrt{2}} \begin{pmatrix} -1 \\ 2 \\ -1 \\ 6 \end{pmatrix}$